

Sumit Srivastava

UX Researcher | Human-AI Interaction & Behavioural Science

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Nationality: Dutch

Summary

Mixed-methods UX researcher with a PhD studying how people understand, trust, and are persuaded by conversational AI. Designs and runs controlled user studies, combines quantitative and qualitative analysis, and measures behavioural outcomes such as trust, persuasion, cognitive load, and comprehension. Grounded in cognitive science and psycholinguistics, with the technical background to prototype the systems being tested and turn findings into design and product decisions.

Research Methods & Skills

Quantitative: Controlled & between-subjects experiments, survey design, statistical analysis (ANCOVA, t-tests, correlation, power analysis), behavioural metrics, NASA-TLX cognitive load

Qualitative: Thematic and qualitative analysis, Wizard-of-Oz studies, and dialogue analysis

Domains: Human-AI interaction, conversational interfaces, explainable AI, trust & persuasion, decision-making

Tools: Python (data analysis), statistical analysis, Rasa / DialogFlow (study prototypes), spaCy

User Research Experience

Doctoral Researcher (User Research, Human-AI Interaction)

University of Twente, Human Media Interaction

2020 – Present

The Netherlands

- Led the full research cycle, from question and study design through data collection, analysis, and reporting, across multiple controlled user studies.
- Investigated how a conversational agent's word choice affects user understanding, trust, and persuasion in collaborative decision-making.
- Built study apparatus (Wizard-of-Oz and LLM-based agents) and recruited, ran, and analysed participant sessions.
- Mentored eight MSc/BSc students through their own research and study design.

M.Sc. Researcher, Artificial Intelligence

KU Leuven

2019 – 2020

Belgium

- Studied how people interpret figurative language, modelling inference with a Rational Speech Act (Bayesian) framework; proposed the topic independently.

Selected Research Projects

- **Trust in human-agent collaborative decision-making** (ACM UMAP, 2024). Between-subjects experiment with a partially wizarded conversational agent on a collaborative decision task; measured trust, agreement ratio, information uptake, and cognitive load (NASA-TLX), analysed with ANCOVA and non-parametric tests.
- **Lexical alignment and comprehension of explanations** (ACM IUI, 2023). Experiment testing whether an agent adapting its word choice changes how well users understand its explanations, using a Cloze-style comprehension measure.
- **Systematic review of alignment measurement** (*Computer Speech & Language*, 2024). Synthesised the existing research into a three-family taxonomy and five evaluation criteria for measuring linguistic alignment in dialogue.

Selected Publications

- Srivastava, S., Theune, M., Catala, A., & Reed, C. (2024). Trust in a Human–Computer Collaborative Task With or Without Lexical Alignment. *ACM UMAP Adjunct '24*.
- Srivastava, S., Theune, M., & Catala, A. (2023). The Role of Lexical Alignment in Human Understanding of Explanations by Conversational Agents. *ACM IUI*, 423–435.
- Srivastava, S., Wentzel, S., Theune, M., & Catala, A. (2024). Measuring and Implementing Lexical Alignment: A Systematic Literature Review. *Computer Speech & Language*, Elsevier.

Industry Experience

Senior Backend Engineer <i>Tibo Energy</i>	2025 – Present <i>The Netherlands</i>
CTO (part-time / freelance) <i>BrainBite; ad:personam GmbH</i>	2021 – 2025 <i>NL / Remote</i>
Technical Lead & Engineer <i>MiQ Digital</i>	2013 – 2019 <i>India</i>

Education

Ph.D., Computer Science (Human–Computer Interaction) <i>University of Twente</i>	2020 – 2026 <i>The Netherlands</i>
• Dissertation: “The Other <i>Wordly</i> Effects: Lexical Alignment in Human-Agent Dialogues” (defence 2026).	
M.Sc., Artificial Intelligence (<i>Cum Laude</i>) <i>KU Leuven</i>	2019 – 2020 <i>Belgium</i>
M.Tech., Information Technology (CGPA 9.2/10) <i>IIT Allahabad</i>	2010 – 2012 <i>India</i>

References

- Dr. Mariët Theune, Associate Professor, Human Media Interaction, University of Twente. m.theune@utwente.nl
- Dr. Alejandro Catala, Assistant Professor, CiTIUS, University of Santiago de Compostela. alejandro.catala@usc.es